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Data Scientist | Machine Learning Engineer

Nairobi, Kenya | +254 112509808 | [Email](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Results-driven AI & Machine Learning Engineer with 2+ years of experience designing and deploying scalable, **end-to-end AI applications** in healthcare, finance, and education. Proven expertise in **fullstack engineering**, **LLM integration**, and **predictive modeling**, consistently translating complex data into production-ready features. Demonstrated track record of **automating workflows**, building **autonomous AI agents**, and deploying intelligent capabilities that reduce operational workloads by up to 40%.

SKILLS

- **AI & Machine Learning:** LLM APIs, AI Agents, Predictive Modelling, Machine Learning, Transformers, Data Science.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, Langchain, LangGraph, FastAPI.
- **Software & Data Engineering:** Python, SQL, Git & Linux, Docker, Data Visualization, Data Analysis, Exploratory Data Analysis.
- **Cloud & Infrastructure:** AWS, Azure, Google Cloud, Snowflake.
- **Statistics:** Hypothesis Testing, Bayesian Statistics, A/B Testing, Statistical Modelling.

WORK EXPERIENCE

AI Applied Engineer

AfyaReach | Nov 2025 - Present

- Architected and deployed **end-to-end AI applications** designed to solve complex healthcare challenges, directly integrating intelligent features into operational platforms.
- Engineered **autonomous AI agents** that successfully automated administrative workflows, drastically reducing manual documentation and doctors' daily workload.
- Built and integrated an LLM-powered backend tool to analyze, align, and maintain medical policies, ensuring **real-time compliance** and **documentation accuracy** for healthcare professionals.

Data Scientist

Data Science East Africa | Jan 2025 - Oct 2025

- Built and deployed production-ready ML models across the healthcare, finance, and education sectors, successfully **reducing workflow loads by 30–40% across domains**.
- Engineered **multimodal AI agents** with contextual conversation capabilities, seamlessly handling text, images, and tabular data to increase user engagement by 40%.
- Optimized data pipelines and automated model retraining workflows, effectively **reducing deployment**

time by 35% and enabling highly scalable, production-scale AI solutions.

Data Scientist

Lutheran World Federation (LWF) | April 2024 – Dec 2024

- Developed highly accurate predictive models to identify at-risk students with **87% accuracy**, enabling proactive interventions that **improved retention rates by 25%**.
- Delivered critical data insights via interactive dashboards and visual reports, **directly influencing \$50K+** in donor funding decisions and guiding education policy adjustments.
- Conducted rigorous program impact evaluations using control groups and regression analysis, isolating program effectiveness with **95% statistical confidence**.

Data Analyst

Lutheran World Federation (LWF) | Sept 2023– March 2024

- Designed and maintained interactive PowerBI dashboards accessed by 10+ stakeholders, accelerating executive decision-making speed by 15%.
- Analyzed large-scale education datasets to uncover operational bottlenecks, driving evidence-based strategies that improved overall program efficiency by 10%.
- Produced visual reports that shaped program management strategies for key policymakers and global donors.

PERSONAL PROJECTS

1. QDOCTOR (GenAI Medical Assistant) [LINK](#)

Developed a NLP powered GenAI solution to help mental health professionals in matters policies that are regularly changed and doctors have to read many documents to keep up to date with them.

Tools: Python, MongoDB, GroqAPI, FastAPI, PyPDF, HuggingFace Transformers, Docker

2. Depression Prediction [LINK](#)

Developed a machine learning pipeline to detect early signs of depression by analyzing behavioral and lifestyle indicators (sleep, diet, financial stress), providing data-driven insights to support timely mental health intervention.

Tools: Python, Pandas, Matplotlib, XGBoost, Random Forest

3. Tomato Crop Disease Detection [LINK](#)

Developed a computer vision model to identify and classify tomato plant diseases, helping smallholder farmers in Kenya reduce crop loss. Achieved **98.05% classification accuracy** on the PlantVillage dataset.

Tools: Python, PyTorch, OpenCV, Kaggle PlantVillage Dataset

EDUCATION

University of Nairobi — Bachelor of Science, Computer Science

LuxDevHQ — Certificate of Data Science, Machine Learning & AI

Deeplearning.AI / Coursera — Mathematics for Machine Learning

LangChain Academy — Building Ambient Agents with LangGraph